

# Universal AI: AIXI

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Iliad Intensive

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- **The common engine**: a single **Bayesian mixture**  $\xi$  over a class  $\mathcal{M}$  of candidate environments, weighted by a prior  $w_\nu$ .
  - Solomonoff:  $\xi$  (eventually) predicts as well as the truth  $\mu$ .
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- **Universal choice**:  $\mathcal{M}$  = all computable environments, prior  $w_\nu = 2^{-K(\nu)}$  ( $K$  is the “complexity” of the environment). Assumption “ $\mu \in \mathcal{M}$ ” becomes “*the universe is computable*”.

# The Bayesian mixture $\xi$

## Definition

Class  $\mathcal{M} = \{\nu_1, \nu_2, \dots\}$  with prior weights  $w_\nu > 0$ ,  $\sum_\nu w_\nu = 1$ .

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- **Key fact (mixture is a predictor):** its one-step prediction is a *posterior-weighted* average of the members' predictions (you prove this today!)

$$\xi(x_t \mid x_{<t}) = \sum_{\nu \in \mathcal{M}} w(\nu \mid x_{<t}) \nu(x_t \mid x_{<t}), \quad w(\nu \mid x_{<t}) = w_\nu \frac{\nu(x_{<t})}{\xi(x_{<t})}.$$



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- **Posterior update is multiplicative:**  $w(\nu \mid x_{1:t}) = w(\nu \mid x_{<t}) \cdot \frac{\nu(x_t \mid x_{<t})}{\xi(x_t \mid x_{<t})}$ : we reweight by how well  $\nu$  predicted vs.  $\xi$  (Bayes' rule!)

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- Spaces: actions  $\mathcal{A}$ , observations  $\mathcal{O}$ , rewards  $\mathcal{R} \subset [0, 1]$ , percepts  $\mathcal{E} = \mathcal{O} \times \mathcal{R}$ , and (finite) histories  $\mathcal{H}^* := (\mathcal{A} \times \mathcal{E})^* := \cup_{t=0}^{\infty} (\mathcal{A} \times \mathcal{E})^t$ .

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- **No Markov / MDP assumption!** Both the policy and environment can depend on all prior percepts.

# The three players

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## Interaction measure $\nu^\pi$

Run  $\pi$  inside  $\nu$  and multiply along the history:

$$\nu^\pi(\mathfrak{a}_{t:m} \mid \mathfrak{a}_{<t}) := \prod_{k=t}^m \pi(a_k \mid \mathfrak{a}_{<k}) \nu(e_k \mid \mathfrak{a}_{<k} a_k).$$



# Value, optimality, and AIXI

## Value function (assuming geometric discount $\gamma$ )

$$\begin{aligned} V_{\nu}^{\pi,m}(\mathfrak{a}_{<t}) &= (1 - \gamma) \mathbb{E}_{\nu}^{\pi}[G_{t-1:m} \mid \mathfrak{a}_{<t}] \text{ with } G_{t-1:m} = \sum_{k=t}^m \gamma^{k-t} r_k \\ &= (1 - \gamma) \sum_{\mathfrak{a}_{t:m}} \nu^{\pi}(\mathfrak{a}_{t:m} \mid \mathfrak{a}_{<t}) \sum_{k=t}^m \gamma^{k-t} r_k \end{aligned}$$

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- **Optimal value**  $V_{\nu}^{*} := \sup_{\pi} V_{\nu}^{\pi}$ ; an optimal policy  $\pi_{\nu}^{*}$  satisfies  $V_{\nu}^{\pi_{\nu}^{*}} = V_{\nu}^{*}$ .
- **Bayes-optimal policy**  $\pi_{\xi}^{*} \in \arg \max_{\pi} V_{\xi}^{\pi}$ : acts optimally w.r.t. the *mixture*  $\xi$ .

## AIXI

$\pi_{\xi}^*$  with  $\mathcal{M}$  = all lower-semicomputable environments and prior  $w_{\nu} = 2^{-K(\nu)}$ . Write  $\xi_U$  for  $\xi$  with this choice of  $\mathcal{M}$  and  $w_{\nu}$ , so  $\text{AIXI} = \pi_{\xi_U}^*$ .

**Intuition:** *The Bayes-optimal agent with prior according to Occam's razor: Act optimally according to environments that you believe to be in, assuming the simplest environments consistent with observations are more likely to be true.*

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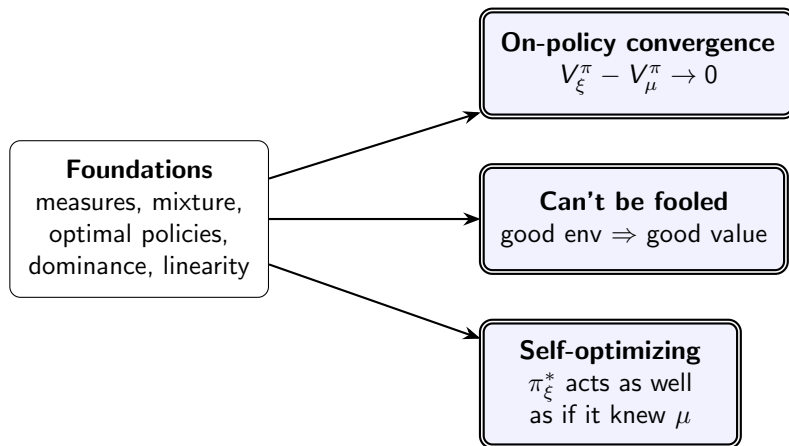
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## AIXI, explicitly

max over actions,  $\sum$  over percepts, choosing actions that maximise the  $\xi$ -weighted discounted return:

$$a_t = \arg \max_{a_t} \lim_{m \rightarrow \infty} \sum_{e_t} \max_{a_{t+1}} \sum_{e_{t+1}} \cdots \max_{a_m} \sum_{e_m} \xi(e_{t:m} \mid \mathfrak{a}_{<t} a_{t:m}) \sum_{k=t}^m \gamma^{k-t} r_k$$

# The three main AIXI results



- All three: the Bayesian agent inherits the guarantees it would have if it knew the truth.

# Result 1: On-policy value convergence

## Theorem

For any  $\mu \in \mathcal{M}$  and any policy  $\pi$ ,

$$V_{\xi}^{\pi}(\mathfrak{a}_{<t}) - V_{\mu}^{\pi}(\mathfrak{a}_{<t}) \rightarrow 0 \quad \mu^{\pi}\text{-almost surely.}$$

**Intuition:** *A policy that acts, pretending the environment is  $\xi$ , will eventually act as well as if it were in the true environment  $\mu$ .*



## Result 2: AIXI can't be fooled

### Theorem (deterministic $\mu$ )

If the truth is reachable-good,  $V_\mu^*(\mathfrak{a}_{<t}) > \varepsilon$  along the played history, then

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**Intuition:** *Whenever good behaviour in  $\mu$  is achievable, the Bayes-optimal agent secures at least a fraction proportional to the prior weight of  $\mu$ .*

## Result 3: Self-optimizing property (stretch goal)

### Theorem

If *some* policy can learn to act optimally in every  $\nu \in \mathcal{M}$ , then  $\pi_\xi^*$  does too:

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**Intuition:** *The Bayes-optimal agent that is unaware of which environment it is in learns to act just as well as the best possible policy that already knows which environment it is in ( $\mu$ ).*

# Takeaways

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- **Why care for safety:** a precise, assumption-light model of idealized intelligence providing a lens on what optimal agents do. For example, we can show under what circumstances AIXI would [wirehead](#) [RO11], [safely rather than recklessly explore](#) [CCH20], or [self-modify](#) [OR11] as a model of what superintelligent agents would do.



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